46110 Fundamentals of Aerodynamics – spring 2026

Assignment 3: Ingenuity Legacy

History was made on April 19, 2021, as Ingenuity, depicted in Figure 1, performed the first powered flight on another planet. It completed 72 flights, greatly surpassing the initial design goal of 5, with over 120 minutes airborne. Sadly, on January 18, 2024, it experienced an emergency landing that broke one of its propellers, ultimately ending its mission.

The new objective is to scale up the payload capacity and flight range for future flights on Mars. Therefore, this project aims to generate the **first conceptual aerodynamic design** for a vehicle that fulfills the new requirements.

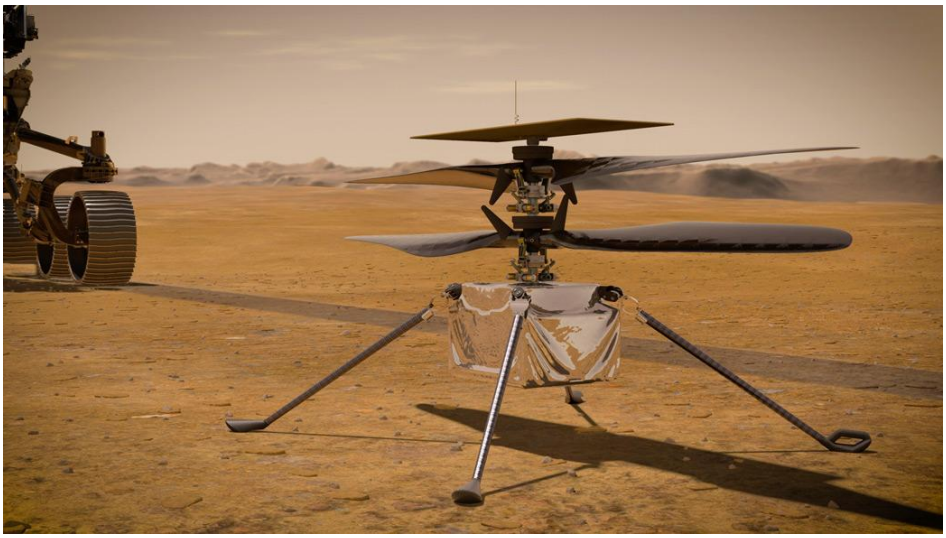

Figure 1 NASA's Ingenuity Mars Helicopter stands on the Red Planet's surface as NASA's Perseverance rover (partially visible on the left) rolls away. Credits: NASA/JPL-Caltech

Assignment format

Deliver a PDF report containing your solution to the tasks and the corresponding code (or a link to its repository). There is no page or word limit, but you are expected to be concise and not repeat theory. You are expected to say what you did, for example, which equation you used to calculate something, explain any assumption you made when implementing a model, show your results, and discuss them by referring to the assumptions from the model or made by you.

You should submit one report per group before the corresponding deadline.

Report Objectives

This assignment aims to assess your ability to implement basic rotor aerodynamic theory in hover and forward flight, apply aerodynamic theory for blade and wing design, and airfoil selection. All of this in the context of a realistic engineering application. Beyond the technical aspects, you will also be evaluated on the clarity of your report, including text and plots.

Task 1 – Ingenuity's power: According to Nasa's website [1,2] and the technology demonstration paper from the Ingenuity project [3], the helicopter had a mass of 1.8 kg, two contra-rotating propellers of 1.2 m in diameter, which rotated between 2300 and 2900 rpm, and the batteries could provide a peak load of 510 W, but only approximately 360 W are required in mid-flight. In hover, these 360 W are used by the rotors, the computer, control motors, and heating systems. Assuming an angular velocity of 2800 rpm, **estimate how much of this power is consumed by each rotor.**

Remember that Mars' atmospheric density and gravity are different from Earth's. State in your document the density and gravity values used and cite the corresponding source. Furthermore, when calculating the losses from the rotors, assume $C_{d0} = 0.02$.

Task 2 – A new Design: Ingenuity's equipment was primarily a camera and terrain sensors, but those were also required to execute the flight, so its payload was zero, and the main purpose of the helicopter was exploration. The new design is meant to carry **2 kg of payload,** and it will have fewer dimension restrictions, so it's allowed to be bigger than Ingenuity, and its rotors do not have to be stacked on top of each other.

In this task, you must evaluate two design options and choose the one you consider best. The two options are:

1. A helicopter with **two** propellers, so similar to Ingenuity, but the rotors are not stacked.
2. A quadcopter.

For each case, you have to find the weight of the aircraft, determine the optimum radius of the propellers, the number of blades, and total power consumption.

For guidance, Figure 2 shows an estimation of the weight distribution of Ingenuity.

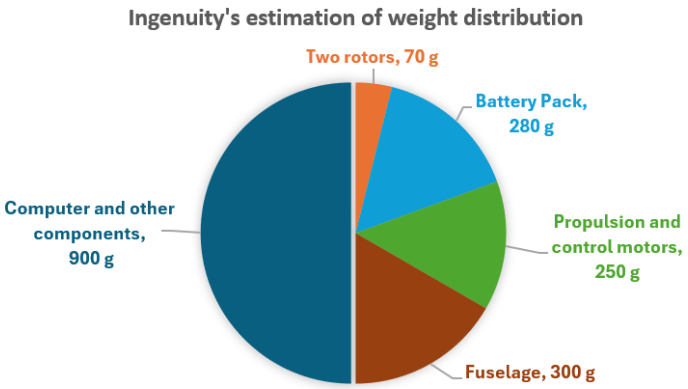

Figure 2. Estimation of weight distribution in Ingenuity. Total weight: 1800 g

Your design also has to comply with the following restrictions and assumptions:

- To have a similar flight time, the battery pack is scaled up and it now weighs 500 g. And they can store 20 Wh of useful energy in flight (it was about 10 Wh for Ingenuity).
- The blades of your propellers are expected to have the same average weight per unit of length as those in Ingenuity. So, the weight per blade in ingenuity is 70g/(2 Rotors*2 blades), and the weight of one of your propellers is $(70\mathrm{g}/4) * \mathrm{N}_{\mathrm{blades}} * (\mathrm{R}/\mathrm{R}_{\mathrm{Ingenuity}})$.

- The weight of the propulsion and control motors of each rotor scales linearly with the power in hover of the rotor (this requires your estimation from task 1). For example, if the propeller of your new design requires a power P_{new}, the weight of the motor driving that propeller is $(250\ g\ /2)P_{new}/P_{Ingenuity}$, where 250g/2 is the weight of the propulsion motors in Ingenuity divided by its number of rotors, and $P_{Ingenuity}$ is the average power consumption per rotor in Ingenuity (estimated in task 1).
- The computer and other components are only slightly larger than in Ingenuity, with a weight of 1000 g.
- The weight of the fuselage scales linearly with the weight of the rest of the aircraft (including the payload). So the mass of the fuselage is $M_{fus} = M_{No-fus} * (M_{fus-Ingenuity}/M_{No-fus-Ingenuity})$

For each design option (two rotors and quadcopter), compare the following in a table:

1. Radius of the rotor
2. Number of blades
3. **Total** power consumption from the rotors (total power required to produce the desired thrust)
4. Aircraft's total mass with payload
5. Flight time in hover

Choose what you believe is the best design and **provide the weight distribution estimation** (like in Figure 2).

Task 3 – Extended battery life: Ingenuity's battery pack has 6 batteries, so each one weighs about 47 g and holds 10/6 Wh of useful flight energy. To extend the flight time, the payload is replaced with more batteries. For your design chosen in Task 2:

- Plot the flight time as a function of the number of extra batteries included and determine the optimum.
- How many extra batteries are the optimum amount, and what is the resulting flight time in minutes?
- If the optimum occurs at a number of batteries larger than what the aircraft can carry (2kg), specify the flight time with the maximum number of batteries that can be carried.

Task 4 – The airfoils: In preparation for the detailed design of the rotors and wings, this task aims to choose appropriate airfoils for the application. To begin with:

- Investigate and report the density and viscosity values of the Mars atmosphere. For the same velocity and chord length, **how much larger or smaller is the Reynolds number on Mars compared to Earth at sea level?**
- Calculate and report the Reynolds number in your rotors at r=0.75 R.

Investigate what airfoil is suitable for this application in the corresponding range of Reynolds numbers and **choose your airfoil with a brief justification** and report:

- The airfoil geometry, and whether it requires or not a turbulator (which forces a turbulent boundary layer).

- Cl vs Cd and Cl vs AoA at the corresponding Reynolds number.

Hint 1: Check the bibliography, such as: *Theory of Wing Sections, Including a Summary of Airfoil Data* [4], and *Airfoil Design and Data* [5]. Airfoil tools website (http://airfoiltools.com/)

Hint 2: In case the source of the airfoil you chose does not have the data at the desired Reynolds, import the geometry and simulate it in Xfoil. However, keep in mind that you may have to increase the number of iterations and use a conservative estimate

Task 5 - Detailed Rotor Design: For the best design obtained in Task 2, determine the geometry of the blades of the rotors. Use the airfoil from the previous task, and determine the following:

- Twist and chord distributions: plot twist in degrees vs non-dimensional radius and chord in meters vs non-dimensional radius.
- Plot the distribution of dC_T ($dC_T = dT/(\rho A V_{tip}^2)$) and dP vs the non-dimensional radius.
- For your final design, calculate the total power required by the rotor and thrust delivered, and compare it with your result from Task 2. Your final design **must** be capable of delivering the desired thrust.

Hint: You have complete freedom to choose any chord and twist distributions (they could, or not, be constant). Different distributions will result in different geometries, and you would like to use the one that provides the required thrust using the minimum power.

Task 6 - Forward flight is achieved by tilting the rotors to generate a horizontal force in addition to the vertical component to hover. On Earth, the power required in this configuration can be greatly reduced if wings are included, but they also add weight. You are tasked to evaluate if it would be worth it to put wings on this drone design for Mars.

Determine the optimum wing configuration to minimize power consumption when carrying a full payload at a ground speed of 10 m/s (assume there is no wind). For calculating the weight of the wings, assume a density similar to the rotor blades, estimated at 74 kg/m^3, and calculate the volume of your wing by integrating the area of the airfoil section along the wingspan. You can also assume that the angle of the propellers concerning the horizontal is independent of the angle of attack of the wing, and the drag coefficient of the body of the helicopter is $C_{D-\,body} = 0.4$.

- Determine your wing shape (airfoil, twist, and chord distributions along the wingspan) and present the plots of **power consumption vs wingspan** and **weight of the aircraft vs wingspan**. Note that a wingspan of b=0 is the case for not implementing a wing.
- The wings, just like the rotor blades, are meant to be rigid and extremely light; they are made of carbon fiber and filled with foam, resulting in an effective density of 74 kg/m^3. However, given the low density of Mars's atmosphere, even these lightweight wings might not be good enough. **Find the density and wing dimensions** required to decrease the power consumption in forward flight by 10%.
- With your final design (with or without wings), plot the total power consumption and rotor angle concerning the horizontal as a function of the flight speed from 0 to 12 m/s.

References:

[1] NASA, (2020), Ingenuity Mars Helicopter, website: https://science.nasa.gov/mission/mars-2020-perseverance/ingenuity-mars-helicopter/#Images

[2] AeroViroment,(2021), MARS HELICOPTER, website: https://www.avinc.com/maccready-works/mars-helicopter

[3] Balaram, B., Canham, T., Duncan, C., Grip, H. F., Johnson, W., Maki, J., Quon, A., Stern, R., & Zhu, D. (2018, January 8). Mars Helicopter Technology Demonstrator. *2018 AIAA Atmospheric Flight Mechanics Conference*. https://doi.org/10.2514/6.2018-0023

[4] Abbott, I.H. and Von Doenhoff, A.E. (1959). Theory of Wing Sections, Including a Summary of Airfoil Data. *Dover Books on Aeronautical Engineering Series*. Dover Publications. ISBN9780486605869

[5] Eppler, R. (1990). Airfoil Design and Data. *Springer Berlin Heidelberg*. ISBN 978-3-662-02648-9. doi: https://doi.org/10.1007/978-3-662-02646-5